COMBINATION VALVE (Continued)

PROPORTIONING VALVE

The proportioning valve is used to balance front-rear brake action at high decelerations. The valve allows normal fluid flow during moderate braking. The valve only controls fluid flow during high decelerations brake stops.

DIAGNOSIS AND TESTING - COMBINATION VALVE

Pressure Differential Switch

(1) Have helper sit in drivers seat to apply brake pedal and observe red brake warning light.

(2) Raise vehicle on hoist.

(3) Connect bleed hose to a rear wheel cylinder and immerse hose end in container partially filled with brake fluid.

(4) Have helper press and hold brake pedal to floor and observe warning light.

(a) If warning light illuminates, switch is operating correctly.

(b) If light fails to illuminate, check circuit fuse, bulb, and wiring. The parking brake switch can be used to aid in identifying whether or not the brake light bulb and fuse is functional. Repair or replace parts as necessary and test differential pressure switch operation again.

(5) If warning light still does not illuminate, switch is faulty. Replace combination valve assembly, bleed brake system and verify proper switch and valve operation.

REMOVAL

(1) Remove brake lines that connect master cylinder to combination valve (Fig. 50).

(2) Disconnect brake lines that connect combination valve to front and rear brakes.

(3) Disconnect wire from combination valve switch terminal. Be careful when separating wire connector as lock tabs are easily damaged if not fully disengaged.

(4) Remove nuts attaching combination valve bracket to booster studs and remove valve bracket off booster studs (Fig. 51).

INSTALLATION

(1) Position valve bracket on booster studs and tighten bracket attaching nuts to 20 N·m (15 ft. lbs.).

(2) Align and start brake line fittings in combination valve and master cylinder by hand to avoid cross threading.

(3) Tighten brake line fittings at combination valve to 19 N·m (14 ft. lbs.).

(4) Tighten brake line fittings at master cylinder to 19 N·m (14 ft. lbs.).

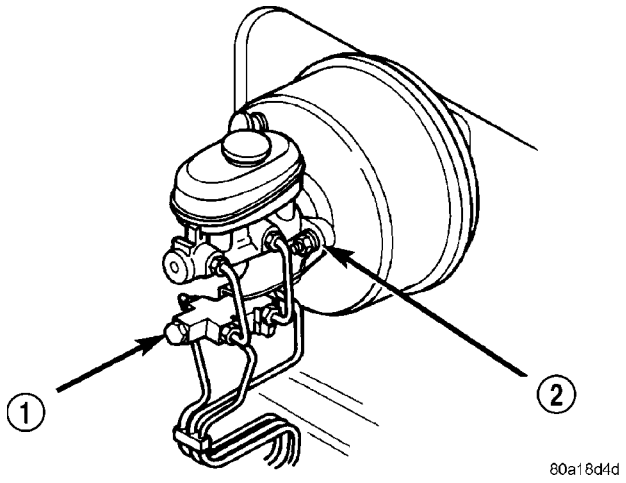

Fig. 50 Combination Valve/Master Cylinder

1 - COMBINATION VALVE
2 - MASTER CYLINDER

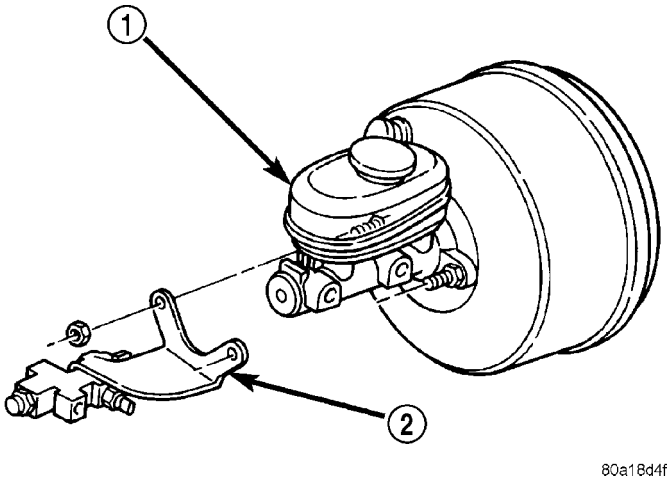

Fig. 51 Combination Valve Bracket

1 - MASTER CYLINDER
2 - COMBINATION VALVE BRACKET

(5) Connect wire to differential pressure switch in combination valve.

(6) Bleed base brake system. (Refer to 5 - BRAKES - STANDARD PROCEDURE).

ROTORS

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - DISC BRAKE ROTOR

The rotor braking surfaces should not be refinished unless necessary.

Light surface rust and scale can be removed with a lathe equipped with dual sanding discs. The rotor